

## ON THE ASYMPTOTIC THEORY OF FARRELL'S SEQUENTIAL CONFIDENCE INTERVAL PROCEDURES

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### SUMMARY

Farrell (Ann. Math. Statist (1966) pp. 586-592) proposed two sequential procedures for obtaining a fixed-width confidence interval for the  $p$ -point of a distribution function. In this paper we study the asymptotic properties of these procedures. A two-stage procedure proposed by Weiss (Naval Res. Logist. Quart. (1960) pp. 251-256) is also considered.

### 1. INTRODUCTION

Let  $F$  be a distribution function having a number  $\gamma_p$  as its unique  $p$ -point ( $0 < p < 1$ ), i.e.  $\gamma_p = x$  is the unique solution of the equation  $F(x) = p$ . By an  $(L, \alpha)$  bounded length confidence interval procedure for  $\gamma_p$  (hereafter simply called an " $(L, \alpha)$  procedure") is meant a sequential procedure  $T$  (based on i.i.d. observations  $X_1, X_2, \dots$  with c.d.f.  $F$ ) with stopping variable  $N$  which upon stopping yields an interval  $[b_N, a_N]$  satisfying

$$(1.1) \quad (i) \quad 0 \leq a_N - b_N \leq L$$

$$(ii) \quad P_F \{ \gamma_p \in [b_N, a_N] \} \geq 1 - \alpha$$

where  $P_F$  is the probability measure corresponding to  $F$ . Should either of the conditions in (1.1) be violated, one says that the given  $(L, \alpha)$  procedure does not work for  $F$ .

Farrell (1964, 1966a, 1966b, to whom the above concept is due) considered estimation of  $\gamma_p$  by  $(L, \alpha)$  procedures and obtained a property of these procedures by making use of results (also derived by him) concerning one-sided sequential hypothesis tests. Defining a class  $D_p^*$  of d.f.'s (which contains all bimodal distributions having continuous

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and strictly positive density functions) and a real-valued functional

$$\epsilon(F) = \sup_{0 < \rho < 1} \min \{F(\gamma_p + \rho L) - p, p - F(\gamma_p - (1-\rho)L)\}$$

(which measures the flatness of  $F$  about its  $p$ -point) on  $D_p^*$ , Farrell (1964; theorem 2) shows that if  $N$  denotes the stopping variable of any  $(L, \alpha)$  procedure that works for *all*  $F \in D_p^*$ , then

$$(1.2) \quad \limsup_{\epsilon(F) \rightarrow 0, F \in D_p^*} EN/g(\epsilon(F)) \geq 2p(1-p)(1-2\alpha), \text{ where } g(x) = x^{-2} \log_2 x^{-2}$$

for  $x \leq e^{-e/2}$

In a subsequent paper (Farrell 1966b) two  $(L, \alpha)$  procedures that work for all  $F \in D_p^*$  are constructed. Denoting by  $N_2$  the stopping variable of the second procedure, Farrell (1966b, section 2) shows that

$$(1.3) \quad \limsup_{\epsilon(F) \rightarrow 0, F \in D_p^*} EN_2/g(\epsilon(F)) \leq 4p(1-p)(1-\alpha/2)$$

(There is an error in the right hand side of this inequality in Farrell (1966b).)

The difference between the right hand sides in (1.3) and (1.2) is  $2p(1-p)(1+\alpha)$ , which is bounded above by  $\frac{1}{2} + \alpha/2$ . This is interpreted by Farrell as indicating that his second procedure is close to being minimax.

From this point on we find it more illuminating to work within a decision – theoretic framework. Let the parameter space  $\Theta$  consist of all sequences  $\{F_n\}_{n=1}^\infty$  of c.d.f.'s in  $D_p^*$  such that  $\epsilon(F_n) \rightarrow 0$  as  $n \rightarrow \infty$ . Thus each point  $\theta \in \Theta$  is in reality a sequence of d.f.'s. Let  $\Upsilon$  denote the set of  $(L, \alpha)$ -procedures that work for every  $F \in D_p^*$ . An element of  $\Upsilon$  is denoted by  $T$  and its stopping variable by  $N_T$ . A risk function is *defined* on  $\Theta \times \Upsilon$  by

$$(1.4) \quad R(\theta, T) = \limsup_{n \rightarrow \infty} EN_T/g(\epsilon(F_n)) \text{ where } \theta = (F_n)_{n=1}^\infty$$

(1.2) is now equivalent to

$$(1.5) \quad \sup_{\theta \in \Theta} R(\theta, T) \geq 2p(1-p)(1-2\alpha) \text{ for every } T \in \Upsilon$$

while (1.3) is equivalent to

$$(1.6) \quad \sup_{\theta \in \Theta} R(\theta, T_2) \leq 4p(1-p)(1-\alpha/2)$$

where  $T_2$  denotes Farrell's second procedure.

As far as we are aware, the equations (1.5) and (1.6) constitute almost all the existing (non-trivial) information about  $R(\theta, T)$ , and the present paper was motivated in the first place by a desire to take a closer look at  $R(\theta, T)$ . In the following  $T_1$  and  $T_2$  are used to denote Farrell's first and second procedures while  $N_1$  and  $N_2$  denote the associated stopping variables.

One question which immediately arises concerns the sharpness of the bound in (1.5). The special structure of  $T_1$  together with careful scrutiny of the proof of (1.5) leads one to suspect that in this case the bound is somewhat conservative. We confirm this by delimiting a subset  $B$  of  $T$  for which it holds that

$$\sup_{\theta \in \Theta} R(\theta, T) \geq 2p(1-p)(1-\alpha) \text{ for all } T \in B$$

and then showing that  $T_1$  is a member of  $B$ .

Next we introduce a third procedure  $T_3$  with stopping variable  $N_3$  satisfying  $N_2 \leq N_3 \leq N_1$ , and show that

$$R(\theta, T_3) = 2p(1-p)(1-\alpha) \text{ for all } \theta \in \Lambda$$

where  $\Lambda$  is a well-defined subset of  $\Theta$ . From this follows

$$\sup_{\theta \in \Lambda} R(\theta, T_2) \leq 2p(1-p)(1-\alpha)$$

which complements (1.6) to some extent. Also one sees that

$$\sup_{\theta \in \Theta} R(\theta, T_3) \geq 2p(1-p)(1-\alpha)$$

which is interesting since there is considerable doubt as to whether  $T_3 \in B$ .

The above results are contained in section 3 of the present paper. To conclude section 3 we consider a two-stage  $(L, \alpha)$  procedure  $T_4$ , with stopping variable  $N_4$ , first proposed by Weiss (1960). Farrell (1966a) has shown that  $T_4$  does not work for all  $F \in D_p^*$ , whereas Weiss (1960) has shown that  $T_4$  does work for all  $F$  having strictly positive continuous and unimodal density functions. Let  $\tilde{D}_p$  denote this class of  $F$ 's and let  $\tilde{\Theta}$  denote the class of all sequences  $\{F_n\}_{n \geq 1}$  in  $\tilde{D}_p$  with  $\epsilon(F_n) \rightarrow 0$ . Then Farrell (1964) has conjectured that the result

$$(1.7) \quad \sup_{\theta \in \tilde{\Theta}} R(\theta, T_4) \geq 2p(1-p)(1-2\alpha)$$

does not hold. Although we cannot verify this conjecture, we have attempted to throw some further light on it by evaluating  $R(\theta, T_4)$  on a subset of  $\Theta$ .

In section 4 we consider the asymptotic behaviour of the four procedures as  $L \rightarrow 0$ . In particular it is shown that

$$\lim_{L \rightarrow 0} \sup E N_2 / g(L) \leq \lim_{L \rightarrow 0} E N_3 / g(L) < \lim_{L \rightarrow 0} E N_1 / g(L),$$

(so that  $T_1$  is in a sense inadmissible), and the asymptotic distribution of  $N_1$  is derived. The exact asymptotic order of  $E(N_4^k)$  is also determined. The results imply that  $E N_4 = o(E N_1)$  as  $L \rightarrow 0$  for certain distributions. This seems to indicate that  $T_4$  is better than  $T_1$  for these distributions. It turns out however that there are good grounds for conjecturing that  $\text{var } N_1 = o(\text{var } N_4)$  as  $L \rightarrow 0$ , which means that  $N_4$  is asymptotically unstable compared to  $N_3$ .

Section 5 considers the asymptotic behaviour of the procedures  $T_1$ ,  $T_2$ , and  $T_3$  as  $\alpha \rightarrow 0$ . To the best of our knowledge very little has appeared in the literature concerning asymptotic theory as  $\alpha \rightarrow 0$  (c.f. Serfling-Wackerley (1976) and Carroll (1976)). We find that there are versions of the three procedures that are essentially equivalent in the sense that

$$\lim_{\alpha \rightarrow 0} E N_i / \log \alpha^{-1} = 2p(1-p) \epsilon(F)^{-2} \text{ for } i = 1, 2, 3.$$

Section 6 considers the construction of versions of  $T_1$ ,  $T_2$  and  $T_3$  that may be used in practical applications. Proofs of all theorems are given in section 7 while section 8 contains a discussion of some open problems.

## 2. Definition of the procedures $T_1$ , $T_2$ , $T_3$ and $T_4$

Let  $X_1, X_2, \dots$  be iid rv's with common c.d.f.  $F$  having a unique  $p$ -point  $\gamma_p$  ( $0 < p < 1$ ). Denote by  $X_{n,1} < \dots < X_{n,n}$  the order statistics of  $X_1, \dots, X_n$  and let

$$(2.1) \quad S_n(\gamma_p) = \sum_{i=1}^n I(X_i \leq \gamma_p)$$

where  $I(\cdot)$  is the indicator function. Let  $\kappa_n^* = \{2n(\log_2 n + c \log_3 n + K)\}^{1/2}$  where  $c > 3/2$  and  $K > 0$  are constants. Applying the Kolmogorov-Feller test (Chung (1968) p 213) we find that  $\{\kappa_n^*, n \geq e\}$  is an upper sequence of  $(S_n(\gamma_p) - np)/\sqrt{p(1-p)}$ . This implies that given  $\alpha > 0$  and an integer  $n_0 > e$  it is possible to choose  $K \equiv h(\alpha)$  in such a way that

$$(2.2) \quad P\{S_n(\gamma_p) \geq np + \{2p(1-p)n(\log_2 n + c \log_3 n + h(\alpha))\}^{1/2} \text{ for some } n \geq n_0\} \leq \alpha/2$$

$$P\{S_n(\gamma_p) \leq np - \{2p(1-p)n(\log_2 n + c \log_3 n + h(\alpha))\}^{1/2} \text{ for some } n \geq n_0\} \leq \alpha/2$$

Note at this stage that  $h(\alpha)$  is in no way determined by  $F$ . In section 6 we discuss methods of calculating  $h(\alpha)$  and  $c$ . We now write, with  $\{\tau_n\}$  denoting a real sequence satisfying  $\tau_n \rightarrow 1$  as  $n \rightarrow \infty$ ,

$$(2.3) \quad \kappa(n) \equiv \kappa(n, \alpha) = \tau_n \{2p(1-p)n(\log_2 n + c \log_3 n + h(\alpha))\}^{1/2} \text{ for } n \geq e.$$

and set, with  $[x]$  denoting the integer part of  $x$ ,

$$(2.4) \quad b(n) \equiv b(n, \alpha) = \max \{1, [np - \kappa(n, \alpha)]\}, a(n, \alpha) \equiv a(n) = \min \{n, [np + \kappa(n, \alpha)] + 1\}.$$

It now follows from (2.4) and (2.2) that

$$(2.5) \quad P\{X_{n, a(n)} > \gamma_p, \text{ all } n \geq n_0\} \geq 1 - \alpha/2$$

$$P\{X_{n, b(n)} < \gamma_p, \text{ all } n \geq n_0\} \geq 1 - \alpha/2$$

Hence, if we define the event

$$(2.6) \quad \Omega_0 \equiv \Omega_{0, \alpha} = \{X_{n, b(n, \alpha)} < \gamma_p < X_{n, a(n, \alpha)} \text{ all } n \geq n_0\}$$

then

$$(2.7) \quad P(\Omega_0) \geq 1 - \alpha$$

For later reference we note that from Farrell (1966b),

$$(2.8) \quad X_{n, b(n)} \rightarrow \gamma_p; X_{n, a(n)} \rightarrow \gamma_p \text{ a.s. as } n \rightarrow \infty$$

In the following  $D_p$  denotes the class of all c.d.f. having a unique  $p$ -point ( $p$  fixed,  $0 < p < 1$ ).

*The procedure  $T_1$*

Let  $N_1 = \text{first } n \geq n_0 \text{ at which } X_{n, a(n)} - X_{n, b(n)} \leq L$ . It follows from (2.8) that  $N_1$  is a.s. well defined. Put  $I_n^{(1)} = [X_{n, b(n)}, X_{n, a(n)}]$ . Then  $T_1$  uses  $I_{N_1}^{(1)}$  as confidence interval for  $\gamma_p$ . It is easy to see that  $T_1$  is an  $(L, \alpha)$  procedure that works for all  $F \in D_p$ .

*The procedure  $T_2$*

Define  $V_n = \max_{n_0 \leq k \leq n} X_{k, b(k)}$  and  $U_n = \min_{n_0 \leq k \leq n} X_{k, a(k)}$  for  $n \geq e$ . Let  $N_2 = \text{first}$

$n \geq n_0$  at which  $U_n - V_n \leq L$ . Since  $N_2 \leq N_1$  it follows that  $N_2$  is a.s. well defined. Put  $I_n^{(2)} = [V_n, U_n]$  if  $V_n < U_n$ , and if  $V_n > U_n$  then  $I_n^{(2)}$  may be taken as any interval of length  $\leq L$ , e.g.  $[\hat{\gamma}_{p,n} - L/2, \hat{\gamma}_{p,n} + L/2]$  where  $\{\hat{\gamma}_{p,n}\}$  is a sequence of estimators of  $\gamma_p$ . Then  $T_2$  uses  $I_{N_2}^{(2)}$  as confidence interval for  $\gamma_p$ .

It is clear that the length of  $I_{N_2}^{(2)}$  is not greater than  $L$ . As to the coverage probability, note that

$P\{\gamma_p \in I_{N_2}^{(2)}\} \geq P\{\gamma_p \in I_{N_2}^{(2)}, \Omega_0\} = P\{\Omega_0\} \geq 1 - \alpha$ , from (2.7). Hence  $T_2$  is an  $(L, \alpha)$  procedure that works for all  $F \in D_p$ .

### *The procedure $T_3$*

Let  $N_2^* = \text{first } n \geq n_0 \text{ at which } V_n > U_n$

$$= +\infty \text{ if } V_n < U_n, \text{ all } n \geq n_0$$

and put  $N_3 = \min\{N_1, N_2^*\}$ . Let  $I_n^{(3)} = I_n^{(1)}$  if  $V_n < U_n$  and e.g.

$$I_n^{(3)} = [\hat{\gamma}_{p,n} - L/2, \hat{\gamma}_{p,n} + L/2]$$

if  $V_n > U_n$ . Then  $T_3$  uses  $I_{N_3}^{(3)}$  as confidence interval for  $\gamma_p$ , and by the same method as for  $T_2$  one can show that  $T_3$  is an  $(L, \alpha)$  procedure that works for all  $F \in D_p$ . Note that  $N_2 \leq N_3 \leq N_1$ .

### *Remark*

Let  $c_n^* \equiv \{2n(\log_2(n+e) + c \log_3(n+e^e))\}^{1/2}$  for  $n \geq 1$ , where  $c > 3/2$ . Then the Kolmogorov-Feller test again shows that  $\{c_n^*, n \geq 1\}$  is an upper sequence of  $(S_n(\gamma_p) - np)/\sqrt{p(1-p)}$ . Since therefore

$$\begin{aligned} & \lim_{m \rightarrow \infty} P\{S_n(\gamma_p) \geq np + \sqrt{p(1-p)} c_n^* \text{ for some } n \geq m\} \\ &= \lim_{m \rightarrow \infty} P\{S_n(\gamma_p) \leq np - \sqrt{p(1-p)} c_n^* \text{ for some } n \geq m\} \\ &= 0, \end{aligned}$$

it follows that if  $\alpha > 0$  is given, there exists a smallest  $m \equiv m(\alpha)$  such

that  $P\{S_n(\gamma_p) \geq np + \sqrt{p(1-p)} c_n^* \text{ for some } n \geq m(\alpha)\} \leq \alpha/2$

and  $P\{S_n(\gamma_p) \leq np - \sqrt{p(1-p)} c_n^* \text{ for some } n \geq m(\alpha)\} \leq \alpha/2$

Let us define

$$(2.9) \quad \alpha(n) = \sqrt{p(1-p)} c_n^* = \{2p(1-p)n(\log_2(n+e) + \log_3(n+e^6))\}^{1/2}, n \geq m(\alpha).$$

Then procedures  $T_1$ ,  $T_2$  and  $T_3$  may also be based on the sequence  $\{\alpha(n), n \geq m(\alpha)\}$  and the asymptotic theory ( $\epsilon(F) \rightarrow 0$  and  $L \rightarrow 0$  with  $\alpha$  fixed) derived in sections 3 and 4 on the basis of the sequence  $\{\kappa(n), n \geq n_0\}$  holds without change if the sequence  $\{\alpha(n), n \geq m(\alpha)\}$  is used. On the other hand, if  $\alpha \rightarrow 0$  while  $F$  and  $L$  remain fixed, it turns out that procedures based on  $\{\kappa(n), n \geq n_0\}$  are incomparably better than those based on  $\{\alpha(n), n \geq m(\alpha)\}$ . Details may be found in section 5.

#### *The procedure $T_4$*

Fix an integer  $m \geq 1$  and write  $Z_i = X_{m+i}$ ,  $i = 1, 2, \dots$

Let  $\gamma_i \equiv \gamma_i(X_1, \dots, X_m)$ ,  $i = 1, 2$  be r.v.'s and  $A$  an event in the  $\sigma$ -field generated by  $X_1, \dots, X_m$  such that

$$(2.10) \quad \gamma_2 < F(\gamma_p + L/2) \text{ and } \gamma_1 > F(\gamma_p - L/2) \text{ on } A$$

Suppose  $P(A)$  is known and let  $v$  be a number such that  $v \cdot P(A) = 1 - \alpha$ .

Define for  $0 < a < b < 1$  and  $n \geq 1$ ,

$$h_n(a, b) = P\{a \leq F(Z_{[np]+1}) \leq b\} = \{n! / ([np])! (n - [np] - 1)!\} \int_a^b y^{[np]} (1-y)^{n-[np]-1} dy$$

and

$$N_W = \text{first } k \geq 1 \text{ at which } h_{n(k)}(\gamma_1, \gamma_2) \geq v$$

where  $\{n(k), k \geq 1\}$  is a sequence of integers satisfying  $n(k) \rightarrow \infty$  as  $k \rightarrow \infty$ .

The two-stage procedure  $T_4$  can now be described.

Let  $I_n^{(4)} = [Z_{[np]+1} - L/2, Z_{[np]+1} + L/2]$ . Stage 1 consists of observing  $X_1, \dots, X_m$ ,

while stage 2 consists of determining  $N_W$  and then using  $I_{n(N_W)}^{(4)}$  as confidence interval

for  $\gamma_p$ . Weiss (1960) has shown that this constitutes an  $(L, \alpha)$  procedure. He has also shown that if  $F$  has an everywhere positive unimodal density function, then  $\gamma_1 = \max \{0, p - \gamma\}$  and  $\gamma_2 = \min \{1, p + \gamma\}$  satisfy (2.10) with

$$A = \{F(X_{m,m}) - F(X_{m,1}) \geq r\}$$

and

$$\gamma = \min \{r - p, r - (1 - p), L \{r - (1 - p)\} / 4 (X_{m,m} - X_{m,1}), L \{r - p\} / 4 (X_{m,m} - X_{m,1})\}$$

where  $r$  ( $0 < r < 1$ ) is arbitrary and  $\{n(k), k \geq 1\}$  is the sequence of positive integers  $n$  for which  $np$  is not an integer.

We write  $N_4 = n(N_W)$  in the sections that follow.

### 3. Asymptotic theory as $\epsilon(F) \rightarrow 0$

Consider a class  $B$  of  $(L, \alpha)$  procedures defined as follows:  $L^{(n)}$  and  $U^{(n)}$ ,  $n = n_0, n_0 + 1, \dots$  are two sequences of r.v.'s which satisfy.

- (3.1) (i)  $L^{(n)} < U^{(n)}$  for  $n \geq n_0$
- (ii)  $L^{(n)}$  and  $U^{(n)}$  are each measurable on  $X_1, \dots, X_n$  for  $n \geq n_0$
- (iii)  $P\{L^{(n)} > \gamma_p \text{ for some } n \geq n_0\} \leq \alpha/2$

and

$$P\{U^{(n)} < \gamma_p \text{ for some } n \geq n_0\} \leq \alpha/2 \text{ for all } F \in D_p.$$

Put  $I_n = [L^{(n)}, U^{(n)}]$   $n \geq n_0$  and use  $I_N$  as confidence interval for  $\gamma_p$  where  $N = \text{first } n \geq n_0 \text{ at which } U^{(n)} - L^{(n)} \leq L$

*Theorem 3.1 (Compare with Theorem 2 of Farrell (1964))*

Let  $D_p^* \subset D_p$  and suppose  $D_p^*$  contains all  $F$  in  $D_p$  having bimodal density functions which are continuous and everywhere positive. Suppose an  $(L, \alpha)$ -procedure  $T \in B$  that works for all  $F \in D_p$  is given and that  $EN < \infty$  for all  $F \in D_p^*$ . Then

$$\sup R(\theta, T) \geq 2p(1-p)(1-\alpha)$$

$$\theta \in \Theta$$

The proof of theorem 3.1 rests essentially on an application of Farrell's (1964) theorem 1 to densities of the form

$$f_q(x) = \begin{cases} q & \text{if } 0 \leq x \leq 1 \\ 1-q & \text{if } 2L \leq x \leq 2L+1 \\ 0 & \text{otherwise} \end{cases}$$

with  $L \geq 1$ . An  $(L, \alpha)$  confidence interval procedure, with terminal interval  $[b_N, a_N]$ , for the  $p$ -point  $\gamma_p$  of such a distribution may be converted into a one-sided hypothesis test of  $q < p$  vs.  $q > p$  as follows:

Decide  $q > p$  if and only if  $a_N \leq 1$ .

In general then, with  $P_{f_q}$  denoting probability under  $f_q$ , for  $q < p$ :

$$P_{f_q} \{ \text{decide } q > p \} = P_{f_q} \{ a_N \leq 1 \} \leq P_{f_q} \{ \gamma_p \notin [b_N, a_N] \} \leq \alpha \text{ and for}$$

$$q > p: P_{f_q} \{ \text{decide } q < p \} \leq \alpha.$$

However, for a procedure in  $B$ , with  $a_N = U^{(N)}$  and  $b_N = L^{(N)}$ , for  $q < p$ :

$$P_{f_q} \{ \text{decide } q > p \} = P_{f_q} \{ U^{(N)} < \gamma_p \} \leq \alpha/2 \text{ and for } q > p: P_{f_q} \{ \text{decide } q < p \} \leq \alpha/2.$$

This serves to explain the increase in the bound obtained in theorem 3.1 over the bound obtained by Farrell.

Since  $T_1 \in B$  we obtain from theorem 3.1.

$$\text{Corollary 3.1} \quad \sup_{\theta \in \Theta} R(\theta, T_1) \geq 2p(1-p)(1-\alpha).$$

It is not known whether  $T_2$  and/or  $T_3$  are members of  $B$ . Nevertheless, by evaluating  $R(\theta, T_3)$  at specific points in  $\Theta$  we are able to show that corollary 3.1 continues to hold if  $T_1$  is replaced by  $T_3$  therein. We shall say that a c.d.f.  $F$  satisfies condition  $A_p$ ,  $0 \leq p \leq 1$ , if  $F$  is everywhere strictly increasing and possesses two bounded derivatives in a neighborhood of its  $p$ -point  $\gamma_p$ . The density of  $F$  is denoted by  $f$  whenever it exists. Finally, we denote by  $\Lambda$  the set of all points  $\theta = \{F_k\}$  in  $\Theta$  such that  $F_k \in A_p$  for

$$k \geq 1 \text{ and } \limsup_{k \rightarrow \infty} \{ f_k(\gamma_p) \cdot \sup_{|x-p| < d} | f'_k(F_k^{-1}(x)) / f^3(F_k^{-1}(x)) | \} < \infty \text{ for some } d > 0.$$

The set  $\Lambda$  is non-empty — take e.g.  $F_k$  to be the uniform distribution on  $[0, k]$  for  $k \geq 1$ . Then the following is obtained:

*Theorem 3.2*

- (i)  $R(\theta, T_3) = 2p(1-p)(1-\alpha)$  for all  $\theta \in \Lambda$
- (ii)  $\sup_{\theta \in \Theta} R(\theta, T_3) \geq 2p(1-p)(1-\alpha)$
- (iii)  $\sup_{\theta \in \Lambda} R(\theta, T_2) \leq 2p(1-p)(1-\alpha)$

It follows from corollary 3.1 and theorem 3.2 that neither  $T_1$  nor  $T_3$  attains Farrell's lower bound (i.e.  $2p(1-p)(1-2\alpha)$ ). The question whether this lower bound is at all attainable remains unanswered.

We conclude this section by considering procedure  $T_4$ . As has already been mentioned, Farrell has conjectured that (1.7) does not hold in this case. Let  $\tilde{\Lambda}$  denote the set of points  $\theta = \{F_k\}$  in  $\Theta$  for which

$$\sup_{k \geq 1} \{f_k(\gamma_p) \sup_{a \leq x \leq b} |f'_k(F_k^{-1}(x))/f_k^3(F_k^{-1}(x))|\} < \infty \text{ for every } 0 < a < b < 1. \text{ Then}$$

*Theorem 3.3*

$$R(\theta, T_4) = 0 \text{ for all } \theta \in \tilde{\Lambda}.$$

Since  $\tilde{\Lambda} \subset \Lambda$ , comparison of theorem 3.3 with theorem 3.2 (i) shows that  $T_4$  is superior to  $T_1$  and  $T_3$ , at least on  $\tilde{\Lambda}$ . (See, however, the remark following theorem 4.3). The available results do not allow for a sensible comparison of  $T_4$  with  $T_2$ .

**4. Asymptotic theory as  $L \rightarrow 0$** 

Throughout this section  $X_1, X_2, \dots$  are i.i.d. with (fixed) continuous c.d.f.F. Before proceeding we state

*Assumption 4.1*

$F$  has two derivatives in some neighborhood of  $\gamma_p$ ,  $F''(x)$  is bounded in the neighborhood and  $F'(\gamma_p) = f(\gamma_p) > 0$ .

Under this assumption we deduce from Bahadur (1966) that

$$(4.1) \quad \sqrt{n/\log_2 n} (X_{n,a(n)} - X_{n,b(n)}) \rightarrow 2\sqrt{2p(1-p)/f(\gamma_p)} \text{ a.s. as } n \rightarrow \infty.$$

*Theorem 4.1*

- (i) As  $L \rightarrow 0$  then  $N_1 \rightarrow \infty$  a.s.,  $N_3 \rightarrow \infty$  a.s. on  $\Omega_0$ .  $N_3 \leq N_2^*$  (independent of  $L$ ) on  $\Omega_0^c$ .
- (ii) As  $L \rightarrow 0$  then  $N_1/g(L) \rightarrow 8p(1-p)/f^2(\gamma_p)$  a.s.  
As  $L \rightarrow 0$  then  $N_3/g(L) \rightarrow I(\Omega_0) \cdot 8p(1-p)/f^2(\gamma_p)$  a.s.
- (iii) The sets  $\{(N_1/g(L))^k\}_{L>0}$  and  $\{(N_3/g(L))^k\}_{L>0}$  are uniformly integrable for every  $k \geq 1$ .

Since from theorem 4.1

$$\lim_{L \rightarrow 0} EN_3/g(L) = P(\Omega_0) \cdot 8p(1-p)/f^2(\gamma_p) < 8p(1-p)/f^2(\gamma_p) = \lim_{L \rightarrow 0} EN_1/g(L)$$

and

$$\lim_{L \rightarrow 0} \text{var} \{N_3/g(L)\} = 0 = \lim_{L \rightarrow 0} \text{var} \{N_1/g(L)\}$$

we see that from an asymptotic expected sample size viewpoint, procedure  $T_3$  performs better than  $T_1$ . Since  $N_2 \leq N_3$  we also have

$$\lim_{L \rightarrow 0} \sup EN_2/g(L) \leq 8p(1-p)f(\gamma_p)^{-2} \cdot P(\Omega_0).$$

We now consider the asymptotic distribution of  $N_1$ . Let

$$T_n = n(X_{n,a(n)} - X_{n,b(n)})/2 \kappa(n)$$

and denote by  $\Phi$  a r.v. having a standard normal distribution.

*Lemma 4.1*

$$f(\gamma_p) \{2\kappa(n)\}^{\frac{1}{2}} (T_n - f(\gamma_p)^{-1}) \xrightarrow{d} \Phi \text{ as } n \rightarrow \infty$$

*Proof:*

This can be derived from results of Siddiqui (1960), c.f. Bloch & Gastwirth (1968: Theorem 2).

In order to derive the asymptotic distribution of  $N_1$  as  $L \rightarrow 0$ , we need to know that Lemma 4.1 remains valid if  $n$  is replaced by  $N_1$  therein. This we have been unable to show. We therefore state the required result as

*Assumption 4.2*

$$f(\gamma_p) \{2\kappa(N_1)\}^{\frac{1}{2}} (T_{N_1} - f(\gamma_p)^{-1}) \xrightarrow{d} \Phi \text{ as } L \rightarrow 0$$

and continue under this assumption. The proof of theorem 4.2 below is also greatly simplified by taking  $\tau_n = 1$  throughout.

Let for  $x \geq e$ ,  $\eta(x) = x(\log_2 x + c \log_3 x + h(\alpha))^{-1}$  with  $c > 3/2$

*Theorem 4.2*

$$f(\gamma_p)^{3/2} \{32p(1-p)\}^{-\frac{1}{2}} L^{3/2} (\log_2 L^{-2})^{-\frac{1}{2}} \{N_1 - \eta^{-1}(8p(1-p)f(\gamma_p)^{-2}L^{-2})\} \xrightarrow{d} \Phi$$

as  $L \rightarrow 0$ .

We conclude the present section by considering the asymptotic behaviour of  $N_4$  as  $L \rightarrow 0$ .

*Theorem 4.3*

(i)  $N_4 < \infty$  a.s. and  $N_4 \rightarrow \infty$  a.s. as  $L \rightarrow 0$ .

(ii) 
$$\lim_{L \rightarrow 0} N_4 L^2 = 16p(1-p) \xi_\alpha^2 (X_{m,m} - X_{m,1})^2 \beta(r,p)^{-2}$$

where  $\beta(r,p) = r - (1-p)$  if  $p \leq \frac{1}{2}$ ;  $= r - p$  if  $p \geq \frac{1}{2}$

and  $(1+\alpha)/2 = \Phi(\xi_\alpha) = \int_{-\infty}^{\xi_\alpha} (2\pi)^{-\frac{1}{2}} e^{-y^2/2} dy$

(iii) If  $E(X_{m,m} - X_{m,1})^{2k} < \infty$  for  $k > 0$ , then the set  $\{N^k L^{2k}\}_{L > 0}$  is uniformly integrable and  $\lim_{L \rightarrow 0} E N_4^k L^{2k} = \{16p(1-p) \xi_\alpha^2 \beta(r,p)^{-2}\}^k E(X_{m,m} - X_{m,1})^{2k}$ .

It follows from the above theorem and from theorem 4.1 that  $EN_i/EN_4 = O(\log_2 L^{-2})$  for  $i = 2, 3$ . This means that from an expected sample size viewpoint  $T_4$  becomes progressively better than e.g.  $T_1$  as  $L \rightarrow 0$ . However, it has recently been shown by Swanepoel and Lombard (1977) that a version of  $T_1$  exists for which  $EN_1/EN_4 = O(1)$  while  $\text{var } N_4/\text{var } N_1 \rightarrow \infty$  as  $L \rightarrow 0$ .

## 5. Asymptotic theory as $\alpha \rightarrow 0$

Referring back to (2.2), (2.3) and (2.4) we take in this section  $n_0 \equiv n_0(\alpha) = \text{first } n \geq e \text{ at which } a(n) < n$ . Also  $\Omega$  will denote the sample space on which the r.v.'s

$X_1, X_2, \dots$  are defined.

*Theorem 5.1*

As  $\alpha \rightarrow 0$  the following statements hold for  $i = 1, 2, 3$ :

- (i)  $N_i \rightarrow \infty$  a.s.
- (ii)  $N_i/h(\alpha) \rightarrow 2p(1-p)\epsilon(F)^{-2}$  a.s.
- (iii) The sets  $\{(N_i/h(\alpha))^k\}_{\alpha > 0}$  are uniformly integrable for each  $k > 0$
- (iv)  $E(N_i/h(\alpha))^k \rightarrow \{2p(1-p)\}^k \epsilon(F)^{-2k}$  for each  $k > 0$ .

*Remarks:*

1. It is easy to show that  $n_0/h(\alpha) \rightarrow 2p/(1-p)$  as  $\alpha \rightarrow 0$ . Hence, from theorem 5.1 (ii) as  $\alpha \rightarrow 0$ ,

$$\frac{N_i}{n_0} \xrightarrow{\text{a.s.}} (1-p)^2/\epsilon^2(F) > 1$$

2. It will be shown in section 6 that there exist sequences  $\{\kappa(n)\}$  satisfying (2.2) and with  $h(\alpha) = O(\log \alpha^{-1})$  as  $\alpha \rightarrow 0$ .

Now let  $N_i^+$  denote the stopping variable of procedure  $T_i$  when it is based on the sequence  $\{c(n), n \geq m(\alpha)\}$  defined in (2.9). We then have

*Lemma 5.1*

$$\liminf_{\alpha \rightarrow 0} \alpha^{1/\epsilon} \log_2 N_i^+ > (\sqrt{\pi\epsilon})^{-1/\epsilon} \text{ a.s. for } i = 1, 2, 3 \text{ where } \epsilon = c - 3/2 > 0.$$

It is now immediate from Theorem 5.1 (ii), remark 2 above and Lemma 5.1 that

$$\liminf_{\alpha \rightarrow 0} \frac{N_i^+}{N_i} = +\infty \text{ a.s.}$$

This shows clearly the advantage of basing the procedures on sequences like the  $\{\kappa(n)\}$  defined in (2.3).

We have not succeeded in discovering the behaviour of  $N_4$  as  $\alpha \rightarrow 0$ . For small values of  $\alpha$  one obtains from the definition:  $h_N(\gamma_1, \gamma_2) \doteq (1-\alpha)P(A)$ . Therefore, if an asymptotic form of  $h_N(\gamma_1, \gamma_2)$  were available, some results could doubtless be obtained.

## 6. Construction of the sequences $\{a(n)\}$ and $\{b(n)\}$

In the present section we briefly consider the construction of sequences  $\{a(n)\}$  and  $\{b(n)\}$  for which (2.5) holds and discuss some aspects relating to practical application of the procedures.

Darling and Robbins (1967: section 2) have shown that

$$\begin{aligned}
 (6.1) \quad P \{S_n(\gamma_p) \geq np + h_n e^{(2\log_2 n)^{-1}} \{2p(1-p)n(\log_2 n + 3\log_3 n + 2,40 + \log \alpha^{-1})\}^{\frac{1}{2}}\} \\
 \text{for some } n \geq 5 \} \\
 = P \{S_n(\gamma_p) \leq np - h_n e^{(2\log_2 n)^{-1}} \{2p(1-p)n(\log_2 n + 3\log_3 n + 2,40 + \log \alpha^{-1})\}^{\frac{1}{2}}\} \\
 \text{for some } n \geq 5 \}
 \end{aligned}$$

$$\leq \alpha/2$$

where  $h_n$  is a known function of  $n$  which satisfies  $h_n \rightarrow 1$  as  $n \rightarrow \infty$ .

Then the  $\kappa(n)$  of (2.3) is fully specified with  $\tau_n = h_n e^{(2\log_2 n)^{-1}}$  and  $h(\alpha) = 2,40 + \log \alpha^{-1}$ .

Let us now consider the case  $p = \frac{1}{2}$  in more detail. Here Darling and Robbins (1967) have shown that the factor  $h_n$  may be dropped without impairing the validity of (6.1). We shall put

$$\kappa^{(1)}(n) = e^{(2\log_2 n)^{-1}} \{(n/2)(\log_2 n + 3\log_3 n + 2,40 + \log \alpha^{-1})\}^{\frac{1}{2}}$$

From a practical point of view, the problem with  $\kappa^{(1)}(n)$  is that the factor  $e^{(2\log_2 n)^{-1}}$  tends to 1 at an extremely slow rate. To illustrate this more clearly, let

$$\kappa^{(2)}(n) = \frac{1}{2} [(n+1)(d^2(\alpha) + \log(n+1))]^{\frac{1}{2}}$$

where  $d = d(\alpha)$  satisfies

$$\exp \{-\frac{1}{2} d^2\} / 2\Phi(d) = \alpha/2$$

e.g. for  $\alpha = 0,05$  we find  $d = 2,4508$  and for  $\alpha = 0,01$ ,  $d = 3,0349$ .

Then from Robbins (1970 p. 1401) we find

$$P \{S_n(\gamma_{\frac{1}{2}}) \geq n/2 + \kappa^{(2)}(n) \text{ for some } n \geq 1\} \leq \alpha/2$$

and similarly

$$P \{S_n(\gamma_{1/2}) \leq n/2 - \kappa^{(2)}(n) \text{ for some } n \geq 1\} \leq \alpha/2$$

Consequently (2.5) continues to hold if  $a(n)$  and  $b(n)$  are defined in terms of  $\kappa^{(2)}(n)$  and the procedures  $T_2$  and  $T_3$  may therefore be based on these “new” sequences  $\{a(n)\}$  and  $\{b(n)\}$ . If we let  $M_2$  and  $M_3$  denote the corresponding stopping variables, then similarly as in theorem 5.1 we can show that

$$\lim_{\alpha \rightarrow 0} EM_2 / \log \alpha^{-1} \rightarrow \epsilon(F)^{-2}/2$$

whence

$$\lim_{\alpha \rightarrow 0} EN_2 / EM_2 = 1$$

This means that asymptotically, as  $\alpha \rightarrow 0$ , there is no advantage in using  $\kappa^{(1)}(n)$  instead of  $\kappa^{(2)}(n)$ .

Since however  $\kappa^{(2)}(n)/\kappa^{(1)}(n) \rightarrow \infty$  as  $n \rightarrow \infty$ , one might argue that only for excessively small values of  $\alpha$  would it hold that  $EN_2 \div EM_2$ , while for “moderate” values of  $\alpha$  one would find  $EN_2 < EM_2$ . The correctness of this argument is immediately in doubt when one looks at the following table of values of  $\kappa^{(1)}(n)$  and  $\kappa^{(2)}(n)$ .

**Values of  $\kappa^{(1)}(n)$  and  $\kappa^{(2)}(n)$  for  $n = 100(100)1000$  and  $\alpha = 0,05; 0,01$**

| n     | $\alpha = 0,05$   |                   | $\alpha = 0,01$   |                   |
|-------|-------------------|-------------------|-------------------|-------------------|
|       | $\kappa^{(2)}(n)$ | $\kappa^{(1)}(n)$ | $\kappa^{(2)}(n)$ | $\kappa^{(1)}(n)$ |
| 100   | 16,38             | 28,55             | 18,68             | 31,14             |
| 200   | 23,84             | 40,20             | 27,01             | 43,69             |
| 300   | 29,69             | 49,17             | 33,50             | 53,35             |
| 400   | 34,68             | 56,74             | 39,04             | 61,50             |
| 500   | 39,13             | 63,41             | 43,96             | 68,68             |
| 600   | 43,17             | 69,45             | 48,43             | 75,17             |
| 700   | 46,91             | 75,01             | 52,56             | 81,15             |
| 800   | 50,41             | 80,18             | 56,42             | 86,71             |
| 900   | 53,72             | 85,04             | 60,06             | 91,94             |
| 1 000 | 56,85             | 89,64             | 63,51             | 96,88             |

Again, it is a simple matter to show that  $\lim_{L \rightarrow 0} EM_3/EN_3 = \infty$ . This result, together with what we see in the table above, leads to the conclusion that the approximation  $EN_3 \doteq g(L)8p(1-p)f(\gamma_{1/2})^{-2}P(\Omega_0)$ , from theorem 4.1, will probably be accurate for purposes of comparison only for rather small values of  $L$ .

Of course, we do not claim that  $\kappa^{(1)}(n, \alpha)$  is the "best possible" sequence satisfying the condition  $\kappa^{(1)}(n, \alpha) \sim \{2p(1-p)n \log_2 n\}^{\frac{1}{2}}$  as  $n \rightarrow \infty$  ( $\alpha$  fixed). We have however, been unable to construct any sequence  $\{\kappa(n)\}$  which satisfies this last condition and which is also a substantial improvement of  $\kappa^{(1)}(n)$ . Taking all this into consideration we would recommend that in practical applications procedure  $T_2$ , based on the sequence  $\{\kappa^{(2)}(n)\}$ , be used.

## 7. Proofs

### Theorem 3.1

We closely follow Farrell's (1964 Theorem 2) proof. Consider the class of c.d.f.'s  $A_p = \{F_q, 0 < q < 1, q \neq p\}$  introduced by Farrell. First we show that the requirement (3.1) (iii) holds also for every  $F_q \in A_p$ .

Let  $A_m = \{(x_1, \dots, x_m): L^{(n)} > \gamma_p \text{ for some } n_0 \leq n \leq m\}$ . Then, with  $P_F$  denoting probability calculated under  $F$ , we find

$$\begin{aligned}
 P_{F_q}(L^{(n)} > \gamma_p \text{ for some } n \geq n_0) \\
 &= \lim_{m \rightarrow \infty} f(A_m) \prod_{i=1}^m f_q(x_i) \\
 &= \lim_{m \rightarrow \infty} f(A_m) \lim_{n \rightarrow \infty} \prod_{i=1}^m f_n(x_i + \rho_n) \text{ from Farrell's equation (17)} \\
 &\leq \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} P_{F_n}(\cdot + \rho_n) \{A_m\} \\
 &\leq \lim_{m \rightarrow \infty} \lim_{n \rightarrow \infty} P_{F_n}(\cdot + \rho_n) \{L^{(n)} > \gamma_p \text{ for some } n \geq n_0\} \\
 &\leq \alpha/2 \text{ since } F_n(\cdot + \rho_n) \in D_p.
 \end{aligned}$$

Similarly we can show that  $P_{F_q}\{U^{(n)} < \gamma_p \text{ for some } n \geq n_0\} \leq \alpha/2$ .

It follows in particular that the given procedure also works for  $F_q \in A_p$ . Letting  $E_{F_q}$  denote expectation under  $F_q$ , we now show that

$$(7.1) \limsup_{|q-p| \rightarrow 0} E_{F_q} N/g(|q-p|) \geq 2p(1-p)(1-\alpha)$$

from which the proof of the theorem will follow along similar lines to those given by Farrell.

Applying procedure T, we obtain a test of  $q < p$  versus  $q > p$  as follows: Decide  $q > p$  if  $U^{(N)} \leq 1$  and decide  $q < p$  if  $L^{(N)} \geq 2L$ .

For this test we obtain

$$\begin{aligned} \text{for } q < p: P_{F_q} \{ \text{decide } q > p \} &\leq P_{F_q} \{ U^{(N)} < \gamma_p \} \text{ since } \gamma_p > 1 \text{ in this case} \\ &\leq P_{F_q} \{ U^{(n)} < \gamma_p \text{ for some } n \geq n_0 \} \\ &\leq \alpha/2 \end{aligned}$$

and for  $q > p$ :  $P_{F_q} \{ \text{decide } q < p \} \leq \alpha/2$ , since  $\gamma_p < 1$  in this case.

As pointed out by Farrell, the above test is really a test about Bernoulli variables  $Z_i$  with  $P \{Z_i = 1\} = q$  for  $i \geq 1$ . Applying Farrell's theorem 1 we find that (7.1) holds.

### Theorem 3.2

The proof of theorem 3.2 is obtained via a series of lemmas. The auxiliary stopping variables  $\tilde{N}_i$  ( $i = 1, 2, 3$ ) to be considered in these lemmas, will now be defined and motivated.

Let  $Y_1, Y_2, \dots$  be a sequence of independent r.v.'s uniformly distributed on  $[0, 1]$  and let  $\tilde{X}_n = F^{-1}(Y_n)$  for  $n \geq 1$  where  $F$  is the common c.d.f. of the r.v.'s  $\{X_n, n \geq 1\}$  and satisfies condition A(p). Recall the definition of the stopping variables  $N_i$  ( $i = 1, 2, 3$ ) (which were defined on the sequence  $\{X_n, n \geq 1\}$ ) and now denote by  $\tilde{N}_i$  the stopping variable on  $\{\tilde{X}_n, n \geq 1\}$  obtained by calculating  $N_i$  on the sequence  $\{\tilde{X}_n, n \geq 1\}$ . Since the sequences  $\{X_n, n \geq 1\}$  and  $\{\tilde{X}_n, n \geq 1\}$  have the same distribution, it follows that  $N_i$  and  $\tilde{N}_i$  also have the same distribution. There is however one very important difference between  $N_i$  and  $\tilde{N}_i$  viz. whereas  $N_i$  is only distributionally dependent on  $F$ ,  $\tilde{N}_i$  is in actual fact functionally dependent on  $F$  (and therefore on  $\epsilon(F)$ ). It follows e.g. that the expression  $\lim_{\epsilon(F) \rightarrow 0} \tilde{N}_3$  is well-defined, while the expression  $\lim_{\epsilon(F) \rightarrow 0} N_3$  is undefined in the usual sense of the word. We also define

$$\Omega_0^* = \{Y_{n,b(n)} < p < Y_{n,a(n)} \text{ for all } n \geq n_0\}$$

and

$$V_n^\star = \max_{n_0 \leq k \leq n} Y_{k,b(k)}; U_n^\star = \min_{n_0 \leq k \leq n} Y_{k,a(k)}$$

*Lemma 3.1*

$$(i) \quad \lim_{\epsilon(F) \rightarrow 0} \tilde{N}_3 = +\infty \text{ on } \Omega_0^\star$$

$$(ii) \quad \tilde{N}_3 < M < +\infty \text{ on } \Omega_0^{\star c} \text{ where}$$

$$M = \text{first } n \geq n_0 \text{ at which } V_n^\star > U_n^\star$$

$$= +\infty \text{ if } V_n^\star < U_n^\star \text{ for all } n \geq n_0$$

*Proof:*

- (i) Let  $r > 0$  be given and put  $\delta(r) = \min \{U_r^\star - p, p - V_r^\star\}$ . Since  $F$  is continuous, it follows from Farrell (1966b p. 592) that there is a  $\rho_0(F) \equiv \rho_0$ ,  $0 < \rho_0 < 1$  such that

$$(7.2) \quad F^{-1}(\epsilon(F) + p) = \gamma_p + \rho_0 L \text{ and } F^{-1}(p - \epsilon(F)) = \gamma_p + (\rho_0 - 1)L$$

Consequently, for  $n = n_0, n_0 + 1, \dots, r$  and any  $F$  with

$$\epsilon(F) < \delta(r): F^{-1}(Y_{n,a(n)}) > F^{-1}(U_r^\star) > F^{-1}(\epsilon(F) + p) = \gamma_p + \rho_0 L,$$

and similarly  $F^{-1}(Y_{n,b(n)}) < \gamma_p + (\rho_0 - 1)L$ . From this it follows that for any

$$F \text{ with } \epsilon(F) < \delta(r): F^{-1}(Y_{n,a(n)}) - F^{-1}(Y_{n,b(n)}) > L, n = n_0, \dots, r$$

$$\text{i.e. } \tilde{N}_3 > r.$$

- (ii) Since  $F$  is strictly increasing, it follows that  $M = \tilde{N}_2^\star$ , where  $\tilde{N}_2^\star$  is  $N_2^\star$  (defined in section 2) calculated on the sequence  $\{\tilde{X}_n, n \geq 1\}$ . On  $\Omega_0^{\star c}$  we have  $M < \infty$  and since  $\tilde{N}_3 \leq \tilde{N}_2^\star = M$  the result follows.

Recall that  $\Lambda$  denotes the set of all points  $\theta = \{F_k\}_{k=1}^\infty$  in  $\Theta$  such that  $F_k \in A(p)$  for

$k \geq 1$  and

$$\limsup_{k \rightarrow \infty} \{f_k(\gamma_p) \sup_{|x-p| \leq d} |f'_k(F_k^{-1}(x))/f'_k(F_k^{-1}(x))| \} < \infty \text{ for some } d > 0.$$

*Lemma 3.2*

If  $\{F_k\}_{k=1}^{\infty} \in \Lambda$ , then  $Lf_k(\gamma_p)/2\epsilon(F_k) \rightarrow 1$  as  $k \rightarrow \infty$ .

*Proof:*

Using  $F_k^{-1}(\epsilon(F_k) + p) = \gamma_p + \rho_0 L$  (see (7.2)) and expanding the left-hand side in a Taylor series yields:

$$\rho_0 \cdot L = \epsilon(F_k)/f_k(\gamma_p) + \frac{1}{2} \epsilon(F_k)^2 \cdot f'_k(F_k^{-1}(\theta_k))/f_k^3(F_k^{-1}(\theta_k))$$

with  $p < \theta_k < p + \epsilon(F_k)$

Similarly, using  $F_k^{-1}(p - \epsilon(F_k)) = \gamma_p + (\rho_0 - 1)L$ , we obtain

$$(\rho_0 - 1)L = -\epsilon(F_k)/f_k(\gamma_p) + \frac{1}{2} \epsilon(F_k)^2 f'_k(F_k^{-1}(\theta_k^{\star}))/f_k^3(F_k^{-1}(\theta_k^{\star}))$$

with  $p - \epsilon(F_k) < \theta_k^{\star} < p$ .

Subtracting the first equation from the second and multiplying by  $f_k(\gamma_p)/\epsilon(F_k)$  we find that for some constant  $C > 0$  and  $k$  sufficiently large

$$|1 - Lf_k(\gamma_p)/2\epsilon(F_k)| \leq \epsilon(F_k) \cdot C \text{ as } k \rightarrow \infty.$$

*Lemma 3.3*

Let  $\{F_k\}_{k=1}^{\infty} \in \Lambda$ . Then

(i)  $\lim_{k \rightarrow \infty} \tilde{N}_3/g(\epsilon(F_k)) = 2p(1-p) \cdot I(\Omega_0^{\star})$ , where  $g(x)$  is defined in (1.2)

(ii) The set  $\{\tilde{N}_3/g(\epsilon(F_k))\}_{k \geq 1}$  is uniformly integrable.

*Proof:*

For notational convenience we write  $N$  in place of  $\tilde{N}_3$  throughout the proof.

(i) A Taylor expansion about the point  $p$  gives, for  $k = 1, 2, \dots$  on  $\Omega_0^{\star}$ :

$$F_k^{-1}(Y_{N,a(N)}) = \gamma_p + (Y_{N,a(N)} - p)/f_k(\gamma_p) + (Y_{N,a(N)} - p)^2 \cdot G_{1,k}$$

and

$$F_k^{-1}(Y_{N,b(N)}) = \gamma_p + (Y_{N,b(N)} - p)/f_k(\gamma_p) + (Y_{N,b(N)} - p)^2 \cdot G_{2,k}$$

where

$$G_{1,k} = \frac{1}{2} f'_k(F_k^{-1}(\phi_1))/f_k^3(F_k^{-1}(\phi_1)), G_{2,k} = \frac{1}{2} f'_k(F_k^{-1}(\phi_2))/f_k^3(F_k^{-1}(\phi_2))$$

and

$$0 < Y_{N,b(N)} < \phi_2 < p < \phi_1 < Y_{N,a(N)} < 1$$

Subtracting the second equation from the first, and using the fact that  $F^{-1}(Y_{N,a(N)}) - F^{-1}(Y_{N,b(N)}) \leq L$  yields for  $k$  sufficiently large and  $C > 0$  some constant,

$$\begin{aligned} L &\geq (Y_{N,a(N)} - Y_{N,b(N)})/f_k(\gamma_p) + (Y_{N,a(N)} - p)^2 G_{1,k} - (Y_{N,b(N)} - p)^2 G_{2,k} \\ &\geq [(Y_{N,a(N)} - Y_{N,b(N)}) + C \{(Y_{N,a(N)} - p)^2 - (Y_{N,b(N)} - p)^2\}]/f_k(\gamma_p) \\ &= (Y_{N,a(N)} - Y_{N,b(N)}) \{1 + C(Y_{N,a(N)} + Y_{N,b(N)} - 2p)\}/f_k(\gamma_p) \end{aligned}$$

Multiplying both sides by  $f_k(\gamma_p)\sqrt{N/\log_2 N} \cdot \tau_N^{-1}$  and using the Bahadur (1966: equations (2), (3)) expansion for uniform r.v.'s gives

$$\begin{aligned} Lf_k(\gamma_p)\sqrt{N/\log_2 N} \tau_N^{-1} &\geq \{2 \cdot \sqrt{2p(1-p)}(1 + c \log_3 N/\log_2 N + h(\alpha)/\log_2 N) \\ &\quad + \tau_N^{-1}\sqrt{N/\log_2 N} R_N\} \cdot \{1 + o(1)\} \text{ as } k \rightarrow \infty \end{aligned}$$

from which it follows, together with lemma 3.2, that

$$\liminf_{k \rightarrow \infty} \epsilon^2(F_k) N/\log_2 N \geq 2p(1-p).$$

It is easily seen that this is equivalent to  $\liminf_{k \rightarrow \infty} N/g(\epsilon(F_k)) \geq 2p(1-p)$  on  $\Omega_0^\star$ . Similarly, using the inequality  $F^{-1}(Y_{N-1,a(N-1)}) - F^{-1}(Y_{N-1,b(N-1)}) > L$ , we find that

$$\limsup_{k \rightarrow \infty} N/g(\epsilon(F_k)) \leq 2p(1-p) \text{ on } \Omega_0^\star. \text{ Therefore } \lim_{k \rightarrow \infty} N/g(\epsilon(F_k)) = 2p(1-p)$$

on  $\Omega_0^\star$ . On the set  $\Omega_0^{\star c}$  we have (lemma 3.1)  $N \leq M < \infty$ , where  $M$  is in no way dependent on  $F_k$ . Consequently  $\lim_{k \rightarrow \infty} N/g(\epsilon(F_k)) = 0$  on  $\Omega_0^{\star c}$ . This completes the proof of (i).

Part (ii) can be demonstrated using the method of Geertsema (1970: Lemma 3.3). We omit the details.

*Lemma 3.4*

- (i).  $R(\theta, T_3) = 2p(1-p)P(\Omega_0)$  for all  $\theta \in \Lambda$
- (ii)  $\sup_{\theta \in \Theta} R(\theta, T_3) \geq 2p(1-p)P(\Omega_0)$
- (iii)  $\sup_{\theta \in \Lambda} R(\theta, T_2) \leq 2p(1-p)P(\Omega_0)$

*Proof:*

Part (i) follows immediately from Lemma 3.3 while (ii) follows from (i). Part (iii) again follows from (i) and the fact that  $N_2 \leq N_3$ .

It is always possible by appropriate choice of  $h(\alpha)$  and by letting  $n_0$  be a finite r.v. (with distribution not dependent on  $F$  in any way) to get  $P(\Omega_0) = 1 - \alpha$ . Since the proof of Lemma 3.3 is unaffected by these changes we see that there exist versions of  $T_2$  and  $T_3$  for which Lemma 3.4 holds with  $1 - \alpha$  replacing  $P(\Omega_0)$ . This completes the proof of theorem 3.2.

*Theorem 3.3*

Let  $Y_1, Y_2, \dots$  denote independent r.v.'s uniformly distributed on  $[0, 1]$  and define  $\tilde{X}_i = F^{-1}(Y_i)$  for  $i = 1, \dots, m$  and  $\tilde{Z}_j = F^{-1}(Y_{m+j})$  for  $j = 1, 2, \dots$

Let  $\tilde{N}_4$  denote  $N_4$  calculated on the sequence  $F^{-1}(Y_i)$ ,  $i \geq 1$ . Let  $\{F_k\}_{k=1}^\infty \in \tilde{\Theta}$ . Since then  $\epsilon(F_k) \rightarrow 0$  through a *unimodal* sequence of  $F$ 's, we easily see that  $\tilde{X}_{m,m} - \tilde{X}_{m,1} \xrightarrow{\text{a.s.}} \infty$ . This implies (taking  $p < \frac{1}{2}$  for definiteness) that  $\tilde{\gamma} = \{r - (1-p)\}L/4(\tilde{X}_{m,m} - \tilde{X}_{m,1})$  for  $\epsilon(F_k)$  sufficiently small. In section 4 it will be shown that if  $\tilde{\gamma} \rightarrow 0$ , then  $\lim \tilde{N}_4 \tilde{\gamma}^2 = \xi_\alpha^2 \cdot p(1-p)$  a.s. where  $(2\pi)^{-\frac{1}{2}} \int_{-\infty}^{\xi_\alpha} e^{-y^2/2} dy = (1+\alpha)/2$ . Hence, in the case under discussion

$$(7.3) \quad \lim_{k \rightarrow \infty} \tilde{N}_4 / (\tilde{X}_{m,m} - \tilde{X}_{m,1})^2 = 16\xi_\alpha^2 p(1-p) L^{-2} \{r - (1-p)\}^{-2} \text{ a.s.}$$

A Taylor expansion at  $p$  gives

$$\tilde{X}_{m,m} - \tilde{X}_{m,1} = (Y_{m,m} - Y_{m,1})/f_k(\gamma_p) + R_m/f_k(\gamma_p)$$

where

$$R_m = f_k(\gamma_p) \{ (Y_{m,m} - p)^2 f'_k(F_k^{-1}(\phi_1)) / f_k^3(F_k^{-1}(\phi_1)) - (Y_{m,1} - p)^2 f'_k(F_k^{-1}(\phi_2)) / f_k^3(F_k^{-1}(\phi_2)) \} / 2$$

with  $|\phi - p| \leq |p - Y_{m,m}|$  and  $|\phi_2 - p| \leq |p - Y_{m,1}|$

Recall that  $\tilde{\Lambda}$  is the set of points  $\theta = \{F_k\}_{k=1}^\infty$  in  $\tilde{\Theta}$  for which it holds that

$$v(\theta, a, b) \equiv \sup_{k \geq 1} \{f_k(\gamma_p) \sup_{a \leq x \leq b} |f'_k(F_k^{-1}(x)) / f_k^3(F_k^{-1}(x))|\} < +\infty \text{ for every } 0 < a < b < 1.$$

Then for every  $\theta \in \tilde{\Lambda}$  we have

$$R_m \leq \frac{1}{2} \{(Y_{m,m} - p)^2 + (Y_{m,1} - p)^2\} v(\theta, p - |Y_{m,1} - p|, Y_{m,m}) = R_m^*$$

Applying this and lemma 3.2 to (7.3) gives

$$\lim_{k \rightarrow \infty} \tilde{N}_4 \epsilon^2(F_k) \leq 4 \epsilon_\alpha^2 p(1-p)^2 (Y_{m,m} - Y_{m,1} + R_m^*)^2 \{r - (1-p)\}^{-2} \text{ a.s.,}$$

and hence  $\lim_{k \rightarrow \infty} \tilde{N}_4 / g(\epsilon(F_k)) = 0$ . If we now assume that limits and expectations are interchangeable, we obtain  $R(\theta, T_4) = 0$  for all  $\theta \in \tilde{\Lambda}$

#### Theorem 4.1

Parts (i) and (ii) follow from the definitions of  $N_1$  and  $N_3$  together with Lemma 1 of Chow and Robbins (1965) and (4.1) above. That part (iii) holds in respect of  $N_1$  can be shown by using the method of proof in Geertsema (1970) Lemma 3.3. Since  $N_3 \leq N_1$ , part (iii) also holds for  $N_3$ .

#### Theorem 4.2

Write  $N_1 = \min(n \geq n_0 : \{n(X_{n,a(n)} - X_{n,b(n)}) / 2\kappa(n)\}^2 \leq (nL / 2\kappa(n))^2)$ .

Then

$$\begin{aligned} & (\kappa(N_1)/2)^{\frac{1}{2}} f(\gamma_p)^2 \{N_1(X_{N_1,a(N_1)} - X_{N_1,b(N_1)}) / 2\kappa(N_1)\}^2 - f(\gamma_p)^{-2} \\ &= (\kappa(N_1)/2)^{\frac{1}{2}} f(\gamma_p)^2 \{N_1(X_{N_1,a(N_1)} - X_{N_1,b(N_1)}) / 2\kappa(N_1) - f(\gamma_p)^{-1}\} \cdot \\ & \cdot \{N_1(X_{N_1,a(N_1)} - X_{N_1,b(N_1)}) / 2\kappa(N_1) + f(\gamma_p)^{-1}\} \xrightarrow{d} \Phi \end{aligned}$$

by (4.1), theorem 4.1 (ii) and assumption 4.2.

It now follows from the theorem of Swanepoel & Geertsema (1976), (by letting  $h(m, L) = (\kappa(m)/2)^{\frac{1}{2}} f(\gamma_p)^2 \{(mL / 2\kappa(m))^2 - f(\gamma_p)^{-2}\}$  there) and from theorem 4.1 (ii) above, that

$$(7.4) \{f(\gamma_p)^{3/2} \{32p(1-p)\}^{-\frac{1}{2}} L^{3/2} (\log_2 L^{-2})^{\frac{1}{2}} \{ \eta(N_1) - 8p(1-p)f(\gamma_p)^{-2} L^{-2} \} \} \xrightarrow{d} \Phi$$

as  $L \rightarrow 0$

Now let

$$\psi(\cdot) = \eta^{-1}(\cdot) \text{ and define } \phi(L) = 8p(1-p)f(\gamma_p)^{-2} L^{-2} \text{ and } \xi(L) = f(\gamma_p)^{3/2} \{32p(1-p)\}^{-\frac{1}{2}} L^{3/2} (\log_2 L^{-2})^{\frac{1}{2}}. \text{ Using (7.4) and a Taylor expansion gives}$$

$$(7.5) P \{ \Phi \leq x \} \stackrel{L}{\rightarrow} P \{ N_1 \leq \psi(x/\xi(L) + \phi(L)) \}$$

$$= P \{ N_1 \leq \psi(\phi(L)) + x\psi'(\phi(L))/\xi(L) + x^2\psi''(\phi(L)) \cdot \theta(L)/2\xi^2(L) \}$$

where  $0 < \theta(L) \rightarrow 1$  as  $L \rightarrow 0$

$$= P \{ \xi(L)(N_1 - \psi(\phi(L))/\psi'(\phi(L))) \leq x + x^2\psi''(\phi(L)\theta(L))/2\psi'(\phi(L)) \cdot \xi(L) \}$$

Straightforward, although rather tedious, computations show that

$$\psi'(\phi(L)) \sim \log_2 L^{-2} \text{ and } \psi''(\phi(L)\theta(L)) = o(L^{3/2}(\log_2 L^{-2})^{-\frac{1}{2}}) \text{ as } L \rightarrow 0.$$

These facts, together with (7.5), establish the theorem.

#### Theorem 4.3

(i) follows from the definition of  $N_4$  and the fact that  $\gamma(L) \rightarrow 0$  as  $L \rightarrow 0$ .

(ii) Using the definition of  $N_4$  and the fact that

$$\lim_{n \rightarrow \infty} \sup_{-\infty < z < \infty} |P \{ d_n \{ p(1-p) \}^{-\frac{1}{2}} (F(X_{[np]+1}) - p) \leq z \} - \Phi(z)| = 0$$

for any sequence of constants  $d_n \sim n^{\frac{1}{2}}$ , we find as  $L \rightarrow 0$ ,

$$\alpha \leq \liminf P \{ \gamma_1 < F(X_{[np]+1}) < \gamma_p | X_1 \dots X_m \}$$

$$= 2 \liminf \Phi(N_4^{\frac{1}{2}} \gamma \{ p(1-p) \}^{-\frac{1}{2}}) - 1, \text{ and}$$

$$\alpha \geq 2 \limsup \Phi(N_4^{\frac{1}{2}} \gamma \{ p(1-p) \}^{-\frac{1}{2}}) - 1$$

Since  $\gamma \equiv \gamma(L) = \beta(r,p)L/4(X_{m,m} - X_{m,1})$  for  $L$  sufficiently small, part (ii) follows.

- (iii) The greater part of the proof of (iii) consists of routine calculations. We shall therefore give only the main points. Let  $L \leq 1$ . Then for each  $x \geq 1$

$$P \{N_4^k L^{2k} \geq x\} \leq P \{\gamma(L) \leq \delta([x^{1/k} L^{-2}], \alpha)\}$$

$$\text{where } P \{|F(X_{[np]+1}) - p| \leq \delta(n, \alpha)\} = \alpha$$

It is easily shown that  $\delta(n, \alpha) < K n^{-\frac{1}{2}}$  for some constant  $K > 0$ . Therefore, and since

$$[x^{1/k} L^{-2}] \geq 1 \text{ for all } L \leq 1 \text{ and } x \geq 1,$$

we have

$$\begin{aligned} P \{N_4^k L^{2k} \geq x\} &\leq P \{\max \{(2K/\beta(r,p))^{2k}, (16K(X_{m,m} - X_{m,1})/\beta(r,p))^{2k}\} \geq x\} \\ &\leq P \{(2K/\beta(r,p))^{2k} + (16K/\beta(r,p))^{2k} (X_{m,m} - X_{m,1})^{2k} \geq x\} \\ &= P \{V \geq x\}, \text{ say} \end{aligned}$$

Since, by assumption,  $EV < \infty$ , the uniform integrability of  $\{N_4^k L^{2k}\}_{L>0}$  follows.

### Theorem 5.1

- (i) follows from the definition of  $N_1$  and the fact that  $n_0(\alpha) \rightarrow \infty$  as  $\alpha \rightarrow 0$ .
- (ii) Let  $0 < \rho < 1$  be such that  $\epsilon(F) = F(\gamma_p + \rho L) - p = p - F(\gamma_p + (\rho-1)L)$ , c.f. Farrell (1966b) p. 592. Now define, following Steyn & Geertsema (1974: p. 31).

$$R_1 = \min(m \geq n_0 : X_{n,a(n)} < \gamma_p + \rho L \text{ for all } n \geq m) \text{ and}$$

$$R_2 = \min(m \geq n_0 : X_{n,b(n)} > \gamma_p + (\rho-1)L \text{ for all } n \geq m)$$

$$\text{Then } N_1 \leq \max(R_1, R_2).$$

From the definition of  $R_1$  it follows that for  $\alpha$  sufficiently small,

$$X_{R_1-1,a(R_1-1)} > \gamma_p + \rho L, \text{ i.e.}$$

$$\sum_{i=1}^{R_1-1} I(X_i \leq \gamma_p + \rho L) < a(R_1 - 1)$$

Since  $R_1 \rightarrow \infty$  as  $\alpha \rightarrow 0$ , it follows from (2.3), (2.4) and the strong law of large numbers that

$$\limsup_{\alpha \rightarrow 0} R_1/h(\alpha) \leq 2p(1-p) \{F(\gamma_p + \rho L) - p\}^{-2}.$$

Similarly we obtain

$$\limsup_{\alpha \rightarrow 0} R_2/h(\alpha) \leq 2p(1-p) \{F(\gamma_p + (\rho-1)L) - p\}^{-2}.$$

Hence

$$(7.6) \quad \limsup_{\alpha \rightarrow 0} N_1/h(\alpha) \leq 2p(1-p) \{F(\gamma_p + \rho L) - p\}^{-2} \text{ a.s.}$$

Now, following Farrell (1966b: p. 592), we consider the r.v.'s

$$N_a(\alpha) = \min(n \geq n_o(\alpha): X_{n,a(n,\alpha)} < \gamma_p + \rho L)$$

and

$$N_b(\alpha) = \min(n \geq n_o(\alpha): X_{n,b(n,\alpha)} > \gamma_p + (\rho-1)L)$$

Using the same type of argument as above we obtain

$$(7.7) \quad \liminf_{\alpha \rightarrow 0} \min \{N_a(\alpha), N_b(\alpha)\}/h(\alpha) \geq 2p(1-p) \{F(\gamma_p + \rho L) - p\}^{-2} \text{ a.s.}$$

Also, for each fixed  $\alpha > 0$ ,

$$N_2(\alpha) \geq \min \{N_a(\alpha), N_b(\alpha)\} I(\Omega_{0,\alpha})$$

and

$$I(\Omega_{0,\alpha}) \rightarrow I(\Omega) = 1 \text{ a.s.}$$

This, together with (7.7) gives

$$(7.8) \liminf_{\alpha \rightarrow 0} N_2(\alpha)/h(\alpha) \geq 2p(1-p) \{F(\gamma_p + \rho L) - p\}^{-2} \text{ a.s.}$$

Combining (7.6) and (7.8), we get the desired result.

(iii) For  $N_1$  the proof is similar to that of Geertsema (1970) Lemma 3.3, with  $h(\alpha)$  replacing  $d^{-2}$ . For  $N_2$  and  $N_3$  the result follows since  $N_2 \leq N_3 \leq N_1$ .

(iv) is a direct consequence of (ii) and (iii).

*Lemma 5.1*

Putting  $\phi(t) = \{2t(\log_2(t+e) + \log_3(t+e))\}^{\frac{1}{2}}$  in theorem 1.4 of Strassen (1967), we find

$$\begin{aligned} \alpha/2 &\geq P \{ (S_n(\gamma_p) - np)/\sqrt{p(1-p)} \geq \phi(n) \text{ for some } n \geq m(\alpha) \} \\ &\sim \int_{m(\alpha)}^{\infty} \phi'(t) \exp \{ -\phi^2(t)/2t \} (2\pi t)^{-\frac{1}{2}} dt \\ &\sim (1/2\sqrt{\pi}) \int_{m(\alpha)}^{\infty} [t(\log t)(\log_2 t)^{c-\frac{1}{2}}]^{-1} dt \text{ since } \phi'(t) \sim (\log_2 t/2t)^{\frac{1}{2}} \\ &= [2\sqrt{\pi} \epsilon (\log_2 m(\alpha))^{\epsilon}]^{-1} \end{aligned}$$

Since, by definition,  $N_1 \geq m(\alpha)$ , the lemma follows.

## 8. Concluding Remarks

In conclusion we briefly indicate some unresolved problems in the theory of the confidence interval procedures considered in this paper.

First and foremost, we still do not know whether the lower bound given in Farrell's (1964) Theorem 2 is attainable or not. In Theorem 3.1 and Corollary 3.1 we have shown that procedure  $T_1$  does not attain Farrell's bound, and in Theorem 3.2 that  $T_3$  (which is closely related to  $T_2$ ) does not either.

Secondly, apart from the upper bound given in section 4 and the properties of the closely related  $T_3$ , very little is known regarding the asymptotic behaviour of  $T_2$  as  $\epsilon(F) \rightarrow 0$  and as  $L \rightarrow 0$ . Finally, regarding asymptotic behaviour as  $\alpha \rightarrow 0$ , procedure  $T_4$  remains to be satisfactorily treated.

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